

MICROMOLD · REQUIREMENTS

MicroMold requirements

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DOCUMENT CONTROL

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Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/micromold		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-09-25	Amish Chadha	First measurable requirements for TRL 2, with status against first-order estimates
0.1	2026-09-25	Amish Chadha	Initial scaffold

MicroMold requirements

These are first-pass requirements for the concept. Targets are proposals for review, not yet validated with users, and will be checked by calculation at TRL 3 and revised after co-design sessions (see MMD-PRB-001). The status column compares each target with the first-order estimates in MMD-PRC-001; "met" means met on paper by an estimate, not demonstrated.

The **reference part** used throughout is a 30 g HDPE test plaque with a 64 x 50 mm projected area, molded from washed, dried flake of 3 to 8 mm.

Table 1. Requirements.

ID	Requirement	Target	Status (estimate)	Verification (TRL 3 or later)
R1	Process the safe recycled thermoplastics	HDPE, PP, LDPE and PS flake of 3 to 8 mm; PVC excluded by procedure and labeling	Met by design	Design review; material trials later
R2	Shot size	30 g or more of HDPE per injection	Met: about 34 g	Swept-volume calculation
R3	Injection pressure	8 MPa (80 bar) or more at the melt with 250 N or less on the handle	Met, thin margin: about 8.9 MPa	Force and friction calculation; later load cell reading
R4	Temperature control	Barrel and nozzle zones each settable from 150 to 260 °C, held within ± 5 °C at the thermocouple; independent cut-out above 280 °C	Met by design, unverified	Controller and heater datasheets; thermal model
R5	Warm-up	Cold start to 220 °C in 15 min or less	Met: about 12 min	Heat capacity calculation
R6	Throughput	8 or more reference parts per hour with one mold	Met on paper: about 10 per hour; melt soak time is the main uncertainty	Cycle-time estimate; later timed trials
R7	Mold envelope	Molds up to 150 x 120 mm footprint and 40 to 120 mm stack height; 40 cm ² or more projected area at full pressure	Met: about 45 cm ² with four M10 bolts	Clamp force calculation
R8	Low-cost tooling	A two-plate aluminum mold for the reference part machinable on a manual mill or small CNC for \$100 or less	Met: about \$90 (indicative)	Quotation from a local shop
R9	Bench size and mass	Press footprint 350 x 300 mm or less, top of handle 1.1 m or less above the bench, mass 35 kg or less without the control box	Met: 320 x 260 mm, about 1.06 m, about 32 kg	Model and mass estimate
R10	Power supply	Single-phase 230 V or 120 V, 1 kW or less, 10 A or less at 120 V	Met: about 600 W, 2.6 A at 230 V, 5 A at 120 V	Heater ratings
R11	Touch-safe outer surfaces	Guard and jacket 60 °C or less at 220 °C set point and 25 °C ambient; nozzle and mold zone guarded	Met on paper for the jacket; nozzle zone guard not yet designed	Thermal calculation
R12	Electrical safety	Earthed frame, fused inlet, double-pole switch, RCD or GFCI supply, heaters and wiring rated for 250 °C at the barrel	Met by design, unverified	Design review against IEC 60204-1 principles
R13	Fume control	Operation only under a local exhaust hood or outdoors; set points above 260 °C blocked in the controller	Not met: no extraction in the concept or BOM	Design review
R14	Affordable	Parts for the press and one mold \$400 or less	Not met: about \$489 (press alone about \$399)	Priced BOM (bom/ bom.csv)

ID	Requirement	Target	Status (estimate)	Verification (TRL 3 or later)
R15	Garage-buildable	Everything except the barrel and the mold built with hand tools, a drill press and optional welding	Partly met: the barrel needs a lathe and the mold a mill, both from a local shop	Build sequence review
R16	Repeatable parts	Part mass within ± 3 % over 10 consecutive shots	Unknown at TRL 2	Later bench trials (TRL 4, not in the current phase)

■ Assumptions

- HDPE melt density about 0.75 g/cm³ at 200 to 220 °C; solid density about 0.95 g/cm³ (typical handbook values, estimates).
- Arbor press drive: pinion pitch radius about 20 mm, handle 450 mm long, overall efficiency of rack, guides and plunger about 60 % (estimates to be checked on the chosen press).
- Processing temperatures follow published melt ranges: HDPE 190 to 230 °C, PP 200 to 240 °C, LDPE 220 to 245 °C, PS 180 to 260 °C ([RJC Mold](#)).
- Four M10 class 8.8 bolts preloaded to about 20 kN each clamp the mold plates, with a safety factor of 2 against melt pressure opening the parting plane.

■ Requirements not met

- **R13 fume control:** the concept has no extraction. Proposed, awaiting Amish: add a small hood with a duct fan (about \$40 to \$60) or make outdoor or hooded operation a written condition of use.
- **R14 cost:** about \$489 with one mold against \$400. See the budget options in MMD-PRC-001.
- **R15 garage-buildable:** partly met; the barrel and molds depend on a local machine shop.
- **R3** is met by a margin of about 10 %, which depends on the assumed drive efficiency.